MILITARY SPECIFICATION

DRILLING MACHINE, RADIAL, FLOOR MOUNTED, 11 INCH X 48 INCH

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1 SCOPE

1.1 Scope. This specification covers a floor mounted, self-contained electric motor driven radial drilling machine 11 inch column diameter and 48 inch arm length. This radial drilling machine shall have a 1 1/2 inch drilling capacity in mild steel.

2 APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DoDISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-152 - Treatment, Moisture- and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment
MIL-V-173 - Varnish, Moisture- and Fungus-Resistant (For Treatment of Communications, Electronic, and Associated Electrical Equipment)
MIL-M-18058 - Machinery, Metal and Wood Working, Packaging of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Center, ATTN: SMCAR-EST-S, Rock Island, IL 61299-7300 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services
FED-STD-376 - Preferred Metric Units for General Use by the Federal Government

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-681 - Identification Coding and Application for Hookup and Lead Wire
MIL-STD-889 - Dissimilar Metals

2.1.2 Other Government documents. The following Government document form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

US DEPARTMENT OF LABOR

OSHA 2206 - General Industry, OSHA Safety and Health Standards (29 CFR 1910)

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted shall be those listed in the issue of the DoDISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN GEAR MANUFACTURERS ASSOCIATION (AGMA)

AGMA 360.02 - Manual for Machining Tool Gearing
AGMA 390.03 - Gear Classification, Materials and Measuring Methods for Unassembled Gears
AGMA 600.01 - AGMA Standard for Metric Usage

(Application for copies should be addressed to the American Gear Manufacturer's Association, One Thomas Circle, Washington, DC 20005.)

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI B5.1 - T-Slots, Their Bolts, Nuts and Tongues
ANSI B5.10 - Machine Tapers, Self-Holding and Steep Taper Series
ANSI B11.8 - American National Standard for Machine Tools - Drilling, Milling, and Boring Machines - Safety Requirements for Construction, Care and Use

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(Application for copies should be addressed to the American National Standards Institute, ATTN: Sales Department, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA-70 - National Electrical Code
NFPA-79 - Electrical Standard for Metalworking Machine Tools and Plastics Machinery

(Application for copies should be addressed to the National Fire Protection Association, 470 Atlantic Avenue, Boston, MA 02210.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informal services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3 REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, a sample shall be subjected to first article inspection (see 4.3, 6.2.1, and 6.4).

3.2 Design. The radial drilling machine shall be new and one of the manufacturer's current commercial models capable of accurately locating and drilling a 1 1/2 inch diameter hole in mild steel to an accuracy true position of not more than 0.0015 inches. The radial drilling machine shall include all components, parts, and features necessary to meet the performance requirements specified herein. The machine shall be so designed that all parts subject to wear, breakage, or distortion shall be accessible for adjustment, replacement and repair with the use of simple hand tools.

3.2.1 Measurement system. The US customary (inch) system of units (US) or the International System of Units (SI) shall be used in the design and construction of the machine. When only one system of measurement is acceptable, the particular system required shall be as specified (see 6.2.1). In this specification, all measurements, dimensions, sizes and capacities are given in the US units. These measurements may be converted to the SI units through the conversion factors and methods specified in FED-STD-376.

3.2.2 Materials. Materials not specifically designated herein or in the contract shall be of quality commensurate with commercial practice within the radial drill manufacturing industry, shall be suitable for the intended purpose in the design of the end item, and shall meet all requirements specified herein. Materials shall be free from defects which would adversely affect the performance or maintainability of the individual components or overall assembly. It shall not be permissible to use reclaimed parts as is, or rebuilt from scrap or other used equipment. When dissimilar

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metals are used in contact with each other, suitable protection against galvanic corrosion shall be applied in accordance with MIL-STD-889.

3.2.3 Energy efficiency. The radial drilling machine and its applicable components that directly comsume energy in normal operation shall be designed and constructed to meet or exceed the energy efficiency of the drilling machine as governed by the latest development available within the drilling machine industry.

3.2.4 Interchangeability. All parts of the end item shall be manufactured to definite standards and tolerances. Parts subject to possible future replacement shall be assigned a manufacturer's part number. When industry or Government standards are applicable, all replaceable parts shall conform to the standards.

3.2.5 Lubrication. All bearings (except sealed for life, permanently lubricated type), mating gears, and all other moving parts shall be provided adequate lubrication by means of a pressurized, self-flushing system. Recirculating systems shall include filters which are cleanable or replaceable. Each lubricant reservoir shall have means for determining fluid levels. All oil holes, grease fittings, and filler caps shall be readily accessible. The supplier shall include and install all lubricants, fluids, greases, and other compounds in accordance with the best commercial practices of the drill manufacturing industry.

3.3 Construction. The radial drilling machine shall be constructed of parts which are new, without defects and free of repairs. The structure shall be capable of withstanding all forces encountered during operation of the machine to its maximum rating and capacity without distortion.

3.3.1 Castings and forgings. All castings and forgings shall be free from defects, scale and mismatching. No process such as welding, peening, plugging or filling with solder or paste shall be used for reclaiming any defective part.

3.3.2 Welding, brazing and soldering. Welding, brazing and soldering shall be of a quality which shall sustain all requirements of the welded, brazed or soldered parts. These operations shall not be employed as repair measures for defective parts.

3.3.3 Fastening devices. All screws, pins, bolts and similar parts shall be installed with means for adjustment and preventing loss of tightness. All parts subject to removal or adjustment shall not be swaged, peened, staked or otherwise permanently deformed.

3.3.4 Surfaces. Surfaces of all castings, forgings, molded parts, stamping and welded parts shall be cleaned and free of sand, dirt, fins, sprues, scale, flux, flush and other harmful or extraneous materials.

3.3.5 Threads. The design, manufacture and configuration of threads used on the machine shall conform to FED-STD-H28 and the applicable "Detailed Standard" section referenced therein.

3.3.6 Dials. Dials shall be graduated in either US customary (inch) unit or SI millimeter (mm) unit. Dial graduations indicating stock removal or tool movement shall be in increments of not more than 0.001 inch or 0.025 mm. Dial diameters shall be such that graduations may be easily read from the machine operator's normal working position. Dials shall be permanently and legibly etched or engraved on a non-glare background. The dials shall be accurate indicators for adjustment of machine or machine component.

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3.3.7 Gears. All gear and pinion design selected for use in the drilling machine or its components shall be manufactured in accordance with diametric pitch (see 6.6.1) conforming to AGMA 390.03 and AGMA 360.02 or metric module (see 6.6.2) system in accordance with AGMA 600.01. All back gears, gears in the headstock area, and gears in the drive train shall be hardened and ground steel.

3.4 Size and capacity. The size and characteristic of the machine shall be not less than those specified in Table I.

3.5 Components.

3.5.1 Base. The base shall be a one piece casting. The top of the base shall be machined for mounting the column and to provide a work surface. The work surface shall have not less than three equally spaced T-slots, 0.75 inch or larger with tie bolts. T-slots shall be in accordance with ANSI B5.1. A coolant fluid trough shall extend around the periphery of the work surface and be connected to a coolant reservoir which shall be integral with the base. A screen to prevent chips from entering the reservoir shall be furnished.

3.5.2 Universal tilting type table. The machine shall have a universal tilting type table configured and machined for attachment to the work surface of the base. The table shall consist of a base on which shall be mounted a tilting work table with two machine-finished work surfaces at 90 degrees to each other. The top surface of the table shall be not less than 20 by 20 inches. The side work surface shall be not less than 10 by 20 inches. The table surfaces shall be machined and provided with T-slots three or more on the top and two or more on the side. Machine T-slots shall be in accordance with ANSI B5.1. The table shall be arranged so that either work surface can be set in a vertical position by means of a worm and segment gear and have means for securing the work surfaces in position. The segment shall be graduated in degrees to indicate the angle at which the table is set. The base of the table shall be finished and provided with slots or bolt holes.

3.5.3 Column. The column shall support the arm, the head, and the spindle and maintain a perpendicularity to the base plate. The trunk and sleeve of the column shall consist of two ferrous castings. The trunk shall be reinforced by internal ribs and shall have a flange for flange base mounting on the base. The column shall be attached to the trunk by means of a clamping device and a 360 degree locking collar totally encompassing the circumference of the clamping devices. The column shall be machined for receiving and mounting the arm. The column shall be void of any cavities, pot holes, crevices, ridges, valleys and peaks. The arithmetical mean deviation (see 6.6.3) of the column surface shall be not greater than 16 microinches.

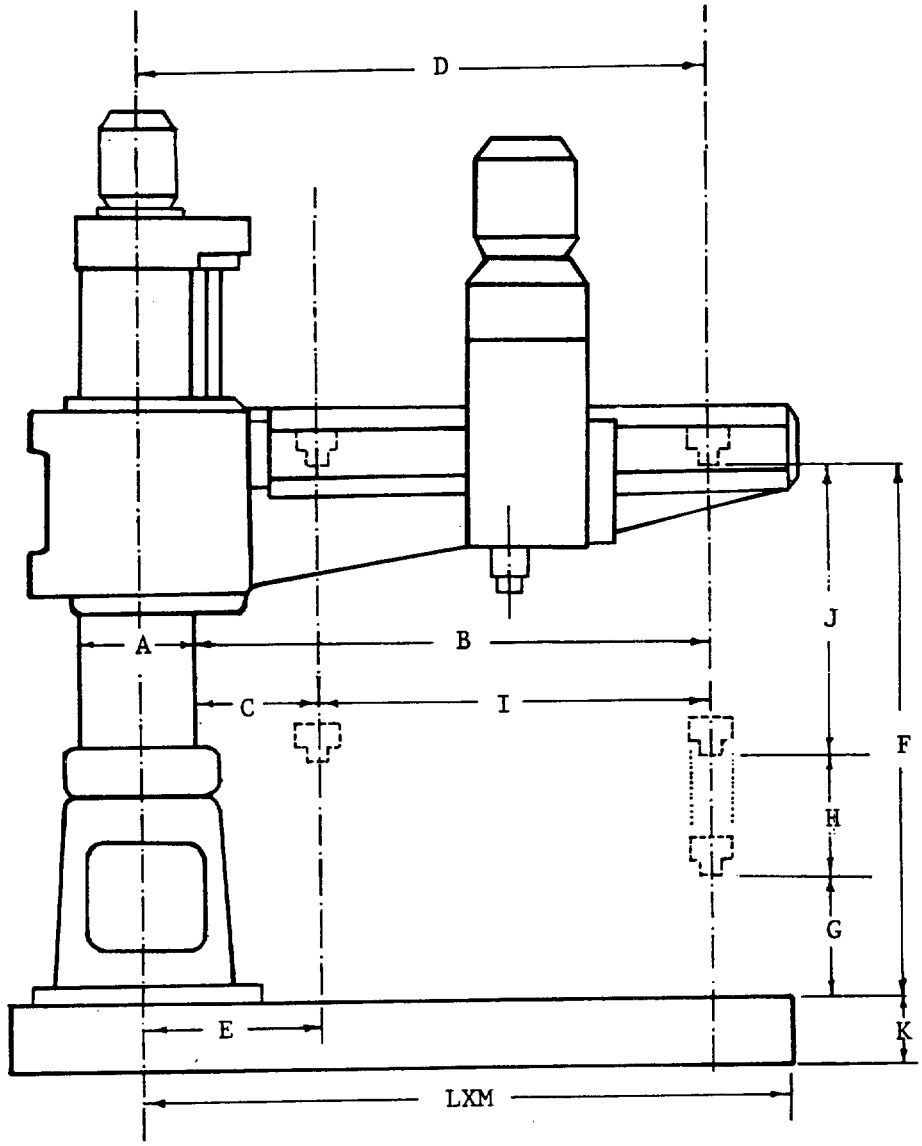
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TABLE I. Size and capacities.

Drilling machine, radial, floor mounted (Dimension in inches unless otherwise specified - see Figure 1)	Size/Capacity
A. Column diameter	11.00
B. Arm length	48.00
C. Column face to center of spindle, minimum	11.00
D. Spindle radius from column center, maximum	55.50 ± 2.00
E. Spindle radius from column center, minimum	16.50
F. Spindle to base, maximum	54.00
G. Spindle to base, minimum, not more than	17.00
H. Spindle traverse in head	12.00
I. Horizontal traverse of head on arm	37.00
J. Arm traverse on column	25.00
K. Base height	8.00
L. Base working length	52.00
M. Base working width	36.00
N. Spindle diameter at point of drive	1.50
O. Spindle speeds, number of	12
P. Spindle speed range, revolution per minute (RPM)	80 to 1500
Q. Spindle feeds, number of	6
R. Spindle speed range, inch per revolution	0.006 to 0.025
S. Taper in spindle, Morse No.	4
T. Motor horsepower, hp	5

Where a range is shown, the requirement is from the stated minimum or less through the stated maximum or greater.

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(See Table I for dimensional values)

FIGURE 1. Drilling machine, radial, floor mounted

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3.5.4 Arm. The arm shall be a ferrous casting, machined and fitted to the column. The length of the arm bearing on the column shall be not less than twice the diameter of the column. Appropriate means shall be provided for lubricating the column sleeve when the arm is raised and lowered. Wipers shall be provided for cleaning the column sleeve. Ways shall be provided for movement of the head. The arm shall have a rack gear for traversing the head along the arm. The arm shall rotate through 360 degrees by means of the column sleeve rotation. Vertical arm movement shall be accomplished by an elevating screw and nut assembly driven either separately or by the main drive motor. The nut assembly shall be mounted on ball or roller bearings and shall be equipped with a safety device that supports the arm in case of thread stripping. Automatic limit stops shall be provided to arrest the travel of the arm at both ends of the vertical range. A clamping device shall be furnished for securing the arm to the column.

3.5.5 Drilling head. The drilling head shall comprise of the spindle drive and feed gearbox, the feed control mechanism, the spindle sleeve, and all operating control members. The head shall be fully enclosed, machined and fitted to the arm. A clamp shall be provided for locking the head to the arm. The head shall be traversed along the arm by means of a power operated rack and pinion gear mechanism. The heads of machines shall be traversed by means of power operated mechanism. The head shall incorporate an interlock or slip clutch to prevent damage from engagement of the rapid traverse while the head is in a locked position. The spindle and all rotating members, shall have sliding surfaces. Sleeves and bearings subject to frictional wear shall be lubricated by a constant supply of oil during use.

3.5.5.1 Spindle. The spindle shall be of heat-treated alloy steel and shall be mounted in a sleeve which shall support or drive the spindle. The spindle shall be supported by ball or roller bearings. The spindle counterbalance mechanism shall be fully enclosed in the head. The spindle nose shall be provided with a Morse taper hole. The spindle taper hole shall be of the size specified in Table I and shall comply with the requirements of ANSI B5.10.

3.5.5.2 Spindle drive mechanism. The spindle drive shall permit not less than 12 spindle speeds. The range of speed shall be as specified in Table I. The speed changes shall be provided by means of sliding heat treated alloy steel gears mounted on multiple splined shafts having not less than three splines with integral keys. The speed changes shall be controlled by not more than three levers and accomplished through sliding gears and positive tooth clutches.

3.5.5.3 Feed control mechanism. The feed mechanism shall provide no less than six feeds. The feed range shall be as specified in Table I. The feed control gears shall be made of heat treated alloy steel and shall slide on multiple splined shafts having not less than three splines with integral keys. A depth gage shall be provided which shall automatically disengage the power feed at any predetermined depth indicated on the graduated dial. A device shall be included to automatically disengage the feed when the limit of spindle travel is reached. The feed engagement shall be operated by a positive tooth or friction type clutch with means for adjustment.

3.5.5.4 Controls. All operating controls shall be mounted on drill head and shall be within normal reach of the operator even with arm in highest position. Feed changes shall be accomplished by means of levers or knobs with a dial micrometer, a plate with permanent or legible characters, so situated that they are clearly visible to the machine operator's normal working position. All controls shall be located so that unintentional operation by normal movement of the machine, operator, or work will be unlikely.

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3.5.6 Coolant system. The coolant recycling system shall include a sump or reservoir, an electrically power driven pump, and all necessary piping. The sump or reservoir shall be of sufficient capacity to permit the continuous flow of coolant while maintaining at least seven gallons of coolant in the sump or reservoir. The system shall have a means for draining and cleaning. The system shall include a strainer which is easily removed for cleaning. A valve and flexible metal hose, conveniently located to the machine operator's normal working position, shall be provided to permit the operator to direct and control the amount of coolant flow.

3.5.7 Hydraulic system. The hydraulic system shall be complete, including all pumps, valves, piping, cylinders and pressure controls. Overpressure protection shall be provided in the high pressure line to prevent damage to components. A filter system shall be provided to insure delivery of clean fluid. The hydraulic reservoir shall have means for determining fluid level.

3.5.8 Electrical system. The electrical system shall conform to applicable requirements of NFPA-79. A disconnect switch shall be provided to disengage all incoming power and shall be separate from the machine's ON and OFF switch. The machine shall be capable of operating on 230/460 volt, 3 phase, 60 hertz power and shall be initially wired to operated on 230 volts. An identified terminal shall be provided suitable for connecting the proper size grounding conductor in accordance with the requirements for the specified power source.

3.5.8.1 Motors. Motors shall be rated for continuous duty and shall have ball or roller bearings of the sealed and permanently lubricated type. Suitable guards or enclosure shall be provided to protect exposed current-carrying parts of motors. Location of motor on the drill shall not be exposed to dripping or spraying oil, water, or other injurious liquid.

3.5.8.2 Control circuit voltage. Auxiliary control circuits shall be isolated from the input electric power supply by a low-voltage transformer having secondary voltage no greater than 115 volts.

3.5.8.3 Electric power supply cable. An electric power supply cable shall be installed on each machine. The electrical power supply cable shall be not less than 1 inch of stripped tinned wire without receptacle plug. The electrical power supply cable shall include proper sheathing in accordance with NFPA-70. The electric power supply cable shall have a voltage and current carrying capacity exceeding that required by the electric power consuming devices, including electric motors as specified in NFPA-70.

3.5.8.4 Color coded electrical circuitry. The wiring in the electrical system shall have individual point to point color coded wires in accordance with system II requirements of MIL-STD-681. The electrical system/schematics provided with each machine shall be an exact duplication of the electrical system and wiring in the radial drilling machine. The diagram/schematic shall clearly identify the individual, point to point, color coded wire and wire termination points in the electrical system.

3.6 Performance.

3.6.1 Operation. The machine shall be capable of performing drilling, tapping, boring and reaming operations for accomplishing all functional characteristics for which it is designed along with those required by Table I. The drilling machine components as specified in 3.5.3 through 3.5.5.4 shall function without error (see 4.6.1).

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3.6.2 Power drilling. The machine shall be capable of performing drilling without any evidence of excess vibration, overheating, chatter, stalling, permanent distortion or deformation of any part of the machine during or after drilling (see 4.6.2).

3.6.3 Alignment and accuracy. The drilling machine shall meet the alignment and accuracy requirement as specified in Table II (see 4.6.3).

3.6.4 Hardness. All back gears, gears in headstock area, and gears in the drive train as specified in 3.3.7 shall be not less than Rockwell C-45. The spindle as specified in 3.5.5.1 shall be not less than Rockwell C-40 (see 4.6.4).

3.6.5 Dielectric strength. The conductors and all electrical components shall withstand, without damage, twice the rated voltage plus 1000 volts root mean square (rms) at 60 hertz (Hz) for a period of 1 minute between mutually insulated circuits, and between ground and insulated circuits (see 4.6.5).

3.7 Ancillary equipment. The following equipment shall be furnished with each radial drilling machine.

a. One set of hand tools normally furnished by the manufacturer.

b. One worklight, adjustable to illuminate the work area, complete with protective shield.

3.8 Lubrication chart or plate. Unless otherwise specified (see 6.2.1), a corrosion-resistant metallic nameplate, with permanently legible characters written in the English language, shall be attached to each machine in a location conveniently visible to the machine operator. If a chart is furnished, it shall be placed in a transparent plastic folder or laminated between clear plastic, permanently sealed sheets with suitable means for mounting. The following information shall be furnished on the chart and plate:

Points of application
Service interval
Type of lubricant
Viscosity
Military or Federal specification number of lubricant

3.9 Nameplate. The nameplate and information plate(s) can be incorporated into one or more corrosion-resistant metallic plate(s) at the machine manufacturer's option. Characters and numerals shall be permanently and legibly engraved, etched, embossed, or stamped in bold face on a contrasting background. The plate shall be secured to the machine with permanent fasteners such as screws, bolts, and rivets. The plate shall be secured with permanent fasteners at each of the four corners. The plate shall include the following information:

Nomenclature
National Stock Number (NSN)
Manufacturer's name
Manufacturer's model number
Manufacturer's serial number (if applicable)
Contract number

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TABLE II. Alignment and Accuracy Requirements.

Geometrical Characteristics of Components 1/	Maximum Allowable Deviation (inch)
Work surface, flat or level in longitudinal direction.	0.0012 per foot
Work surface, flat or level in transverse direction.	0.0012 per foot
Column square with base in plane through center line of base (sleeve inclined towards front side) for machines up to 4 feet 7 inches length of radial arm.	0 to 0.0024 per foot
Column square with base in plane perpendicular to plane through center line of base plate.	0.0012 per foot
Arm parallel with base (dipping at outer end) for machines up to 4 feet 7 inches length of radial arm.	0 to 0.0024 per foot
Spindle head guide flat or level.	0.0012 per foot
Head movement on arm parallel to base (arm set parallel to the direction of the longitudinal axis of the base, arm locked)	0.012 over 40 inch measuring length
Radial movement of arm parallel to base at three positions of head equally spaced along movement of arm (head locked for each of its three positions).	0.002 over 12 inch measuring length
Runout of the internal taper of the spindle (spindle retracted):	
a. At the mouth of taper (arm and head locked).	0.001
b. At a distance of 12 inch from the spindle nose arm and head locked).	0.002
Spindle axis square with base plate (arm and head locked prior to taking measurements). 2/	0.0024 per foot
Travel of spindle sleeve square with base in plane through center line of base (inclined at lower end towards column)	0.004 per foot

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TABLE II. Alignment and Accuracy Requirements (continued)

Geometrical Characteristics of Components	Maximum Allowable Deviation (inch)
Travel of spindle sleeve square with base in plane 90 degrees to above plane.	0.002 per foot
Drill spindle square with plain type table mounted on base in plane through center line of base (spindle inclined at lower end towards column).	0 to 0.0024 per foot
Drill spindle square with plain type table mounted on base in plane 90 degrees to above plane.	0.0012 per foot

1/ Unless specified, arm positioned halfway up the column and head halfway along the arm.
2/ Arm positioned in its upper position, in its mid-travel position and then in its lower position.

3.10 Protective finish. The protective finish, unless otherwise specified (see 6.2.1), shall be painted. All surfaces to be painted shall, immediately prior to painting, be cleaned and dried and free of all foreign matter. The protecting paint coating shall be at least 2.0 MIL in thickness. The paint adhesion shall be such that no paint shall be capable of being peeled by fiber glass tape applied over two intersecting cuts of not less than two inches long and down to the base metal. The surface hardness of the paint shall be between the industrial standards of H to HB pencil lead. Where no protective finish is utilized (such as bedways, machine guide surfaces, etc.) the natural finish of the material or the finish obtained from heat treatment is permissable provided the surfaces are free from scale or corrosion (see 4.6.6).

3.11 Safety and health requirements. Covers, guards, or other safety devices shall be provided for all parts of the machine that present safety hazards. The safety hazards shall not interfere with the operation of the machine. The safety devices shall prevent unintentional contact with the guarded part and shall be removable to facilitate inspection, maintenance and repair of the parts. All machine parts, components, mechanisms, and assemblies furnished on the machine, whether or not specifically required herein, shall comply with ANSI B11.8 and with all of the requirements of OSHA 2206 that are applicable to the machine itself.

3.12 Fungus control. Components of electrical circuitry that are hermetically sealed or inherently resistant to fungus growth and moisture need not be treated. Components of electrical circuitry that are not inherently resistant to fungus growth or moisture shall be treated in accordance with MIL-T-152 and MIL-V-173.

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3.13 Workmanship. The quality of workmanship imparted to the drilling machine and all its components shall be equal to or exceed the quality of the current commercial drilling machine manufactured by domestic producers of the type of drilling machine addressed herein. Where the requirements herein exceed commercial requirements the requirements herein shall take precedence. The neatness, safety, sharp edges, structual integrity (seams, laps, steps, etc.), material soundness (no cracks, deformations, structual weakness, etc.), brazing and soldering, painting, welding, etc. of the drilling machine shall equal or exceed the workmanship requirements as stated herein. The quality of workmanship shall be such as to eliminate degradation of the form, fit, function, performance and appearance as specified herein.

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3)

b. Quality conformance inspection (see 4.4)

4.3 First article inspection. When required, first article inspection shall be applied to the preproduction model or initial production item (see 3.1). Unless otherwise specified (see 6.2.1), first article inspection shall consist of the examination in 4.5 and all tests under 4.6. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.4 Quality conformance inspection. Quality conformance inspection shall be applied to production items offered for acceptance under the contract. Unless otherwise specified (see 6.2.1), quality conformance inspection shall consist of (a) through (h) below. Failure of any item to pass an examination or test shall be cause for rejection of the item.

a. Product examination (see 4.5)

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b. Operational tests (see 4.6.1)

c. Power drilling test (see 4.6.2)

d. Alignment and accuracy test (4.6.3)

e. Hardness test (see 4.6.4)

f. Dielectric strength test (see 4.6.5)

g. Paint test (see 4.6.6)

h. Packaging inspection (see 4.7)

4.4.1 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105 using inspection levels and acceptable quality levels (AQL) as listed.

Inspection or Test	Inspection Level	AQL
a. Product examination (4.5)	100	
b. Operational test (4.6.1)	II	1.0
c. Power drilling test (4.6.2)	S-4	1.0
d. Alignment accuracy test (4.6.3)	II	1.0
e. Hardness test (4.6.4)	S-4	1.0
f. Dielectric strength test (4.6.5)	S-4	1.0
g. Paint test (4.6.6)	S-4	1.0
h. Packaging inspection (4.7)	S-4	4.0

4.5 Product examination. Visually and manually examine the drilling machine to determine conformance with the requirements of 3.2 through 3.5.8.4, 3.7 through 3.9, and 3.11 through 3.13. Visual examination shall include verification of completeness of manufacture and assembly, conformance to specified standards, adequacy of markings, proper cleaning, and freedom from identified defects. Manual examination shall include the operation of movable parts by hand to assure proper functioning. The examination provisions may be applied at the earliest practical point in manufacture at which it is feasible to inspect for acceptance without risk of change in the characteristic by subsequent operations. Failure of the supplier to provide objective evidence that the drilling machine has passed the examinations prescribed for it by the supplier's inspection system shall be cause for rejection.

4.6 Tests. Measuring instruments used in performing the following tests shall be more accurate by a factor of ten than the specified limit, and shall be tracable to the National Bureau of Standards.

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4.6.1 Operational test. The machine shall be operated at no load for not less than 30 minutes. During the test, the radial arm shall be driven to the maximum upper and lower positions to verify operation of the automatic limit stops. The drilling head interlock or slip clutch shall be tested by attempting to drive the drilling head while in the locked position. The spindle speed shall be varied from 80 rpm to 1500 rpm in no less than 12 steps. The spindle feed rate shall be varied from 0.006 inch/rev to 0.025 inch/rev in no less than six steps. Failure of any control, motor, adjustable mechanism or accessory to perform in accordance with design requirements specified herein shall be cause for rejection (see 3.6.1).

4.6.2 Power drilling test. The machine shall be subjected to a drilling test which applies not less than half the rated input power to the spindle motor for a least one minute. The lowest practical spindle speed shall be chosen by the supplier for this test without applying torque in excess of the maximum design value of the machine components. Material and dimensions of the workpiece, tooling, depth of cut, and rate of feed rate shall be at the option of the supplier to achieve the required loading. Any evidence of chatter, stalling, or permanent distortion during or after the test shall be cause for rejection (see 3.6.2).

4.6.3 Alignment and accuracy test. The machine and its components shall be tested for conformance with the alignment and accuracy as specified in paragraph 3.6.3. Failure to meet alignment and accuracy tolerance requirements shall be cause for rejection (see 3.6.3).

4.6.4 Hardness test. Hardness tests shall be performed on test samples of the spindle, back gears, and all gears in the head stock and drive train. The test samples shall be of the same material and finish as the components and shall have been processed at the same time. Failure of a test sample to meet the hardness as specified in 3.6.4 shall be cause for rejection. A certificate of conformance (COC) may be accepted for a component in lieu of the test, provided the COC is signed by a responsible officer of the original equipment manufacturer of that component.

4.6.5 Dielectric strength test. The electrical circuits shall be subjected to a 60 Hz test voltage of 1000 volts rms plus twice the rated input voltage. The test voltage shall be applied between mutually insulated circuits, and between each conductor and the machine frame. The voltage shall be increased at a minimum rate of 500 volts per second to the test voltage level, held there for one minute, and then removed at the same rate. During the one minute of test voltage, the circuitry shall be observed for flashover (surface discharge), sparkover (air discharge) and breakdown (puncture discharge). Any evidence of flashover, sparkover, breakdown, or damage to the unit due to heat or other effects of the applied voltage shall be cause for rejection (see 3.6.5).

4.6.6 Paint test. The hardness, adhesion and thickness of the paint used on the machine shall be checked as follows:

a. The graphite point of a sharpened HB lead pencil shall be pressed at a right angle into a painted area of the machine until the graphite crumbles. Failure of the surface to withstand the test, as evidenced by a pitting of the paint, shall be cause for rejection.

b. A test sample of the same metal, painted with the same process and at the same time as the machine, shall have a sharpened H lead pencil pressed into the paint as in (a). Failure of the surface to pit (paint too hard) shall be cause for rejection.

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c. Two 2 inch intersecting lines shall be cut through the paint to the base metal of the test sample. A strip of fiberglass tape shall be firmly pressed over the intersection and then pulled from the surface. Adhesion of any paint to the tape shall be cause for rejection.

d. Using a paint thickness measuring instrument, or by masking a portion of the test sample during painting, measure the thickness of the paint. Failure of the paint to be at least 0.002 inch thick shall be cause for rejection (see 3.10).

4.7 Packaging inspection. Packaging inspection shall be conducted before and after packaging to determine compliance with the preservation, packing and marking requirements of Section 5.

5 PACKAGING

5.1 Military preservation and packing. When specified in the contract, requirements for level A preservation, and levels A or B packing shall be in accordance with MIL-M-18058.

5.2 Marking. When military levels of preservation and packing are specified, marking shall be in accordance with the requirements of MIL-STD-129.

5.3 Military and industrial packing for lubrication systems. The lubrication systems shall be temporarily capped or plugged to prevent loss of lubricants during shipment. Tags, warning that all temporary caps or plugs must be removed before start-up or use of the drilling machine, shall be attached to all temporary caps or plugs. These warnings shall also be included in the instruction manual.

5.4 Industrial preservation, packing and marking. When specified in the contract, preservation, packing and marking shall be in accordance with ASTM D3951. Bar coding is also required.

5.4.1 Additional industrial requirements. In addition to the requirements of ASTM D3951, all unprotected ferrous metal parts and all accessories shall be cleaned, preserved and wrapped with greaseproof wrap. Accessories shall be packed, blocked and cushioned in fiberboard container(s). Any critical surfaces shall be protected from damage in transit. Technical data shall be placed in a plastic bag (stapled or sealed closed) and enclosed in accessory containers. Movable parts of the machine shall be secured by mechanical devices incorporated into the machine for that purpose. They must also be augmented with appropriate wood blocking, metal strapping, etc., to assure immobilization. Levers, cranks, handles, etc., shall be secured in position with annealed wire or steel strapping and protected from damage. All items shall be protected from marring.

6 NOTES

6.1 Intended use. The radial drilling machine covered by the specification is intended for drilling, tapping, boring, and reaming operations in mild steel.

6.2 Ordering data.

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6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. If the drilling machine is required to be configured in either US or SI systems, state required system (see 3.2.1).

c. When first article is required for inspection and approval (see 3.1, 4.2 and 6.4).

d. Lubrication chart or plate, if different (see 3.8).

e. Protection finish, if different (see 3.10).

f. First article inspection, if different (see 4.3).

g. Quality conformance inspection, if different (see 4.4).

6.3 Contract data requirements. Required technical data such as operator's manuals, parts lists, wiring diagrams, and examination, original alignment and tolerance test reports and other instructions for operation and maintenance (as identified on a DD Form 1664, Data Item Description) should be specified on a DD Form 1423, Contract Data Requirements List incorporated into the contract.

6.4 First article. When a first article inspection is required, the item should be the first article offered for acceptance under the contract. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article.

6.5 Repair parts manuals. Repair parts and accessory identification in repair part manuals should include the prime contractor's name, nomenclature and part number. Repair part identification in repair part manuals should also include the original equipment/part manufacturer's (OEM) (actual part manufacturer's) name, Federal Supply Code Number (FSCM), nomenclature and part number of the part used in the end item, immediately after the prime contractor's part identification. Items with high mortality rates should be so identified. See 3.1.2.2 through 3.1.2.10 (especially 3.1.2.8.2.2) of MIL-M-7298C.

6.6 Definitions.

6.6.1 Diametral pitch. The ratio of the number to the pitch expressed in inches (or the quotient of the number of teeth by the reference diameter expressed in inches) (see 3.3.7).

6.6.2 Module. The ratio of the pitch, expressed in millimeters, to the number (or the quotient of the reference diameter, expressed in millimeters, by the number of teeth) (see 3.3.7).

6.6.3 Arithmetical mean deviation. The arithmetical mean of the absolute values of the profile departures within the sampling length (see 3.5.3).

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